Arquitectura de Robots

Homogeneous matrix

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Geometric Interpretation of Homogeneous Matrixes

As already described, a homogeneous matrix is used to transform a vector expressed in homogeneous coordinates regarding a system B, to its expression in the coordinates of the reference system A.

It can also be used to rotate and turn a vector referred to a fixed reference system, and ultimately it is used to express the orientation and position of a reference system B with respect to another fixed reference system A.

Geometric Interpretation

The transformation matrix T is usually written as follows:

$$T = \begin{bmatrix} n_x & o_x & a_x & P_x \\ n_y & o_y & a_y & P_y \\ n_z & o_z & a_z & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{n} & \mathbf{o} & \mathbf{a} & \mathbf{p} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $\mathbf{n}, \mathbf{o}, \mathbf{a}$ constitutes an orthonormal basis delineating the orientation and $\mathbf{p}$ denotes a vector specifying the position.

Geometric Interpretation orthonormal configuration

Thus, the vectors **n**, **o**, and **a** delineate a right orthonormal configuration, signifying that:

$$\begin{gathered} ||\mathbf{n}|| = ||\mathbf{o}|| = ||\mathbf{a}|| = 1 \\ \mathbf{n} \times \mathbf{o} = \mathbf{a} \end{gathered} \tag{1}$$

The rotation submatrix **[n,o,a]** corresponds to an orthonormal matrix, which satisfies the following:

$$\begin{bmatrix} \mathbf{n} & \mathbf{o} & \mathbf{a} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{n} & \mathbf{o} & \mathbf{a} \end{bmatrix}^{T} \tag{2}$$

Geometric Interpretation inverse matrix

The inverse matrix of the homogeneous transformation matrix T is easily obtained and corresponds to the following expression:

$$T^{-1} = \begin{bmatrix} n_x & n_y & n_z & -\mathbf{n}^T\mathbf{p} \\ o_x & o_y & o_z & -\mathbf{o}^T\mathbf{p} \\ a_x & a_y & a_z & -\mathbf{a}^T\mathbf{p} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

If we have the relation ${}^{A}P = T{}^{B}P$, and we multiply both sides by T^{-1}, we have:

$$T^{-1}\ {}^{A}P = {}^{B}P \tag{4}$$

Composition of the homogeneous matrix

A complex transformation can be decomposed into a sequential application of elementary transformations, specifically fundamental rotations and translations.

For example, a matrix representing a rotation by angle a on the X_A axis, followed by a rotation by angle α on the Y_A axis and a rotation by angle θ on the Z_A axis, can be obtained by the composition of the basic rotation matrices.



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Rotation composition

Rotation matrices can be composed to express the continuous application of several rotations.

Consequently, should frame A undergo a rotation by an angle α about the X-axis, succeeded by a rotation of angle ϕ about the Y-axis, and culminating in a rotation by angle θ about the Z-axis, the resultant composite rotation can be articulated as follows:

$$T = R(Z,\theta)R(Y,\phi)R(X,\alpha) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & -s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$$

Rotation composition cont...

$$T = R(Z,\theta)R(Y,\phi)R(X,\alpha) = \begin{bmatrix} c\theta c\phi & -s\theta c\alpha + c\theta s\phi s\alpha & s\theta s\alpha + c\theta s\phi c\alpha \\ s\theta c\phi & c\theta c\alpha + s\theta s\phi s\alpha & -c\theta s\alpha + s\theta s\phi c\alpha \\ -s\theta & c\phi s\alpha & c\phi c\alpha \end{bmatrix}$$

where $c\theta$ expresses $\cos\theta$ and $s\theta$ expresses $\sin\theta$

It is important to consider the order in which the rotations are performed, since the matrix product is not commutative.

Rotation composition cont...

If the initial rotation is performed by an angle θ about the Z-axis, succeeded by a rotation by an angle ϕ about the Y-axis, and culminating with an additional rotation by an angle α about the X-axis, the resultant total rotation can be described as follows:

$$T = R(X,\alpha)R(Y,\phi)R(Z,\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix} \begin{bmatrix} c\theta & 0 & -s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathrm{T} = \mathrm{R(X},\alpha)R(Y,\phi)R(Z,\theta) = \begin{bmatrix} c\alpha c\theta & -c\phi s\theta & s\phi \\ s\alpha s\phi c\theta + c\alpha s\theta & -s\alpha s\phi s\theta + c\alpha c\theta & -s\alpha c\phi \\ -c\alpha s\phi c\theta + s\alpha s\theta & c\alpha s\phi s\theta + s\alpha c\theta & c\alpha c\phi \end{bmatrix}$$

Section 2.1

Example

Example

2 Rotation composition

It is necessary to obtain the transformation matrix that represents the $\{B\}$ system obtained from the $OX_AY_AZ_A$ system by means of a rotation of angle -90 around the OX_A axis, a translation of vector $p(5, 5, 10)$and a rotation of 90ř on the OZ_A axis.

To do this, it will be enough to multiply the different basic matrices in the appropriate order:

$$T = T(z, 90)T(p(x, y, z))T(x, 90) =$$

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 & -5 \\ 1 & 0 & 0 & 5 \\ 0 & -1 & 0 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Section 2.2

Example 2

Example

2 Rotation composition

Obtain the transformation matrix that represents the following transformations on a fixed $\{A\}$ reference system: translation of a vector $p_{x,y,z}(-3, 10, 10)$; rotation of 90ř about the OX_B axis of the translated system and rotation of 90ř about the OY_B axis of the rotated system.
The corresponding basic matrices are chosen and composed in the reverse order of the previous example.

$$T = T(p(x_A, y_A, z_A))T(x_B, -90)T(y_B, 90) =$$

$$\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & -3 \\ -1 & 0 & 0 & 10 \\ 0 & -1 & 0 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In general

2 Rotation composition

In general, when composing various transformations using homogeneous matrices, the following criteria must be considered.

1. If the fixed system $OX_AY_AZ_A$ and the transformed system $OX_BY_BZ_B$ are coincident, the homogeneous transformation matrix will be the 4×4 identity matrix, $I_{4\times4}$.

In general

2 Rotation composition

1. In the scenario where the coordinate system $OX_BY_BZ_B$ is derived through a series of rotations and translations from the fixed reference frame $OX_AY_AZ_A$, the associated homogeneous transformation matrix corresponding to **each operation must be pre-multiplied** by the matrices representing the preceding transformations.
2. If the coordinate frame $OX_BY_BZ_B$ is derived through sequences of rotations and translations with respect to the mobile reference frame, the homogeneous transformation matrix corresponding to each individual operation **must be postmultiplied** by the matrices representing the preceding transformations.

Premultiply and Postmultiply

2 Rotation composition

In linear algebra, the terms ”premultiply.ând ”postmultiplyrefer to how matrices are multiplied in relation to other matrices or vectors. The difference lies in the order of multiplication:

1. **Premultiply**: - **To premultiply a matrix** means to multiply it by another matrix on the left. That is, if you have a matrix A and you multiply it by another matrix B from the left, you are premultiplying B by A. - Mathematically, this is expressed as $C = A \cdot B$, where A premultiplies B.

2. **Postmultiply**: - **To postmultiply a matrix** means to multiply it by another matrix on the right. In this case, if you have a matrix A and you multiply it by another matrix B from the right, you are postmultiplying A by B. - Mathematically, this is expressed as $C = B \cdot A$, where B postmultiplies A.

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Q&A

Thank you for your attention

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